Chapter 12 • Information Handling

Solved Exercise 12.1

1. The chemistry score distribution is 24–28:3, 29–33:6, 34–38:12, 39–43:23, 44–48:15, 49–53:6. Answer the eight class questions.

(i) What is the upper limit of the last class? The last class is 49–53, so its upper limit is 53.
(ii) What is the lower limit of class 39–43? It is 39.
(iii) What is the midpoint of 34–38? $\frac{34+38}{2}=36$.
(iv) What are the frequencies of 29–33 and 44–48? They are 6 and 15 respectively.
6 and 15
(v) What is the class size? For example, $29-24=5$, so the class size is 5.
(vi) Which class has the minimum number of students? The smallest frequency is 3, in 24–28.
(vii) What is the lower limit of the class whose frequency is 15? That class is 44–48, so the lower limit is 44.
(viii) How many students scored between 24 and 43? Add the first four frequencies: $3+6+12+23=44$.
44 students.

2. The chair-repair expenditures (rupees in hundred) are: 145,152,153,156,158,160,146,152,155,159; 161,163,165,147,148,151,154,156,158,160; 144,167,151,150,152,149,145,153,152,155. Prepare eight classes by the tally-bar method and give the last three frequencies.

Minimum=144, maximum=167, range=23. With eight classes, use class width 3: 144–146, …, 165–167.
Tally each observation once. The successive frequencies are 4,3,7,5,4,4,1,2 and their total is 30.
Therefore the last three classes 159–161,162–164,165–167 have frequencies 4,1,2 respectively.
Last three class frequencies: 4, 1, 2.

3. The weights (kg) of 30 students are: 30,33,24,21,15,39,37,44,42,33; 33,28,29,32,31,28,26,32,34,35; 38,36,41,30,35,41,23,26,18,34. Using six classes, prepare a frequency table and construct a frequency polygon.

Minimum=15, maximum=44, range=29. Six classes of width 5 give 15–19 through 40–44.
Counting the observations gives frequencies 2,3,5,10,6,4, whose total is 30.
Class midpoints are 17,22,27,32,37,42; join these frequency points and extend to zero at the exterior midpoints 12 and 47.
Frequencies=2,3,5,10,6,4; total=30.

4. The Grade-10 English marks are: 58,59,58,33,40,58,45,46,43,45,45; 50,52,49,50,57,52,55,49,50,62,49; 48,44,42,47,46,47,46,53,40,44. Use five classes, find the class with least frequency and construct a histogram.

Minimum=33, maximum=62, range=29. Use width 6: 33–38,39–44,45–50,51–56,57–62.
Counting each mark once gives frequencies 1,6,15,4,6 respectively; their total is 32.
Thus 33–38 has the least frequency, 1. Draw adjacent bars using boundaries 32.5–38.5, …, 57.5–62.5.
Least-frequency class=33–38, frequency 1; total=32.

5. Draw a frequency polygon on the histogram for weights 10–14 through 35–39 with frequencies 6,17,23,30,22,13.

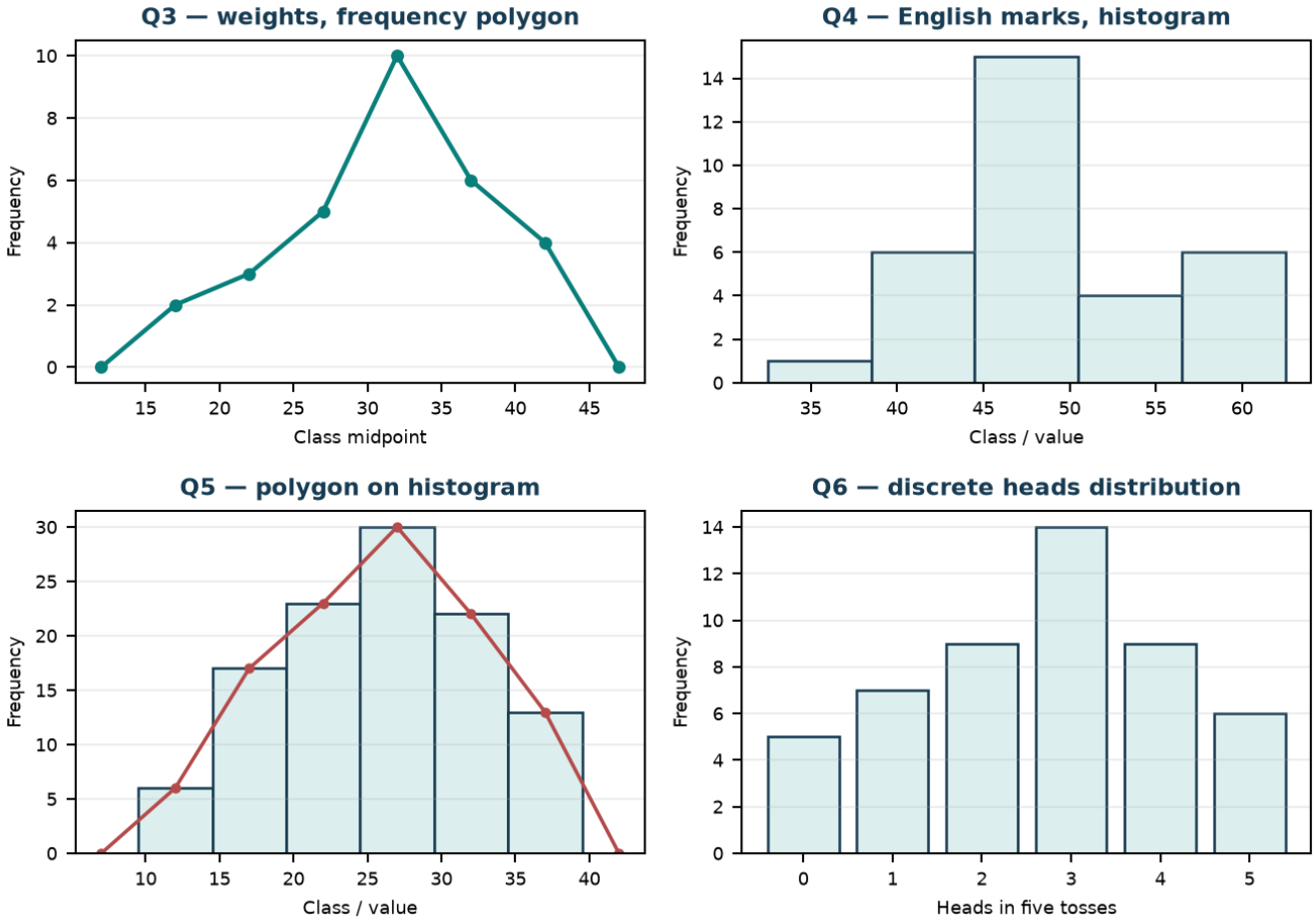
Use boundaries 9.5–14.5, …, 34.5–39.5. Join bar-top midpoints 12,17,22,27,32,37 and extend to zero at midpoints 7 and 42.

6. Fifty five-coin experiments produced these numbers of heads: 3,3,4,0,5,4,3,3,1,2,4,5,0,3,2,4,4,0,0,0,5,5,3,2,1; 4,3,2,5,3,2,1,3,5,4,3,2,1,3,2,1,3,1,3,1,4,3,2,2,4. Make a discrete frequency distribution.

Count every occurrence of 0,1,2,3,4,5 separately.

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The counts are respectively 5,7,9,14,9,6. Their sum is 5+7+9+14+9+6=50, as required.
Frequencies 0–5 = 5,7,9,14,9,6 respectively.



7. Draw the histogram for marks 35–37,38–44,45–54,55–61,62–67,68–72 with frequencies 2,12,16,13,9,3.

Class boundaries: 34.5–37.5,37.5–44.5,44.5–54.5,54.5–61.5,61.5–67.5,67.5–72.5.
Widths = 3,7,10,7,6,5. Frequency densities
= f/width≈0.667,1.714,1.600,1.857,1.500,0.600.

8. Make a frequency polygon on the histogram for continuous classes 5–8,8–12,12–20,20–25,25–27,27–32 with frequencies 2,12,25,32,14,5.

Widths=3,4,8,5,2,5
densities≈0.667,3,3.125,6.4,7,1.
For the polygon on this unequal-class histogram, join bar-top midpoint densities at 6.5,10,16,22.5,26,29.5 and extend to zero.

